
DETERMINISTIC MEMORY FUSION FOR LONG-HORIZON CONVERSATIONAL QA: CROSS-MODEL EVIDENCE BEYOND LLM-GUIDED FUSION

ABSTRACT

To maintain factual coherence across long conversational sessions, memory-augmented agents must continuously consolidate redundant dialogue fragments into compact, retrievable representations — a process known as memory fusion that accounts for the dominant share of system performance, with its removal collapsing multi-hop F1 by over 53%. However, fusion universally relies on additional LLM calls at maintenance time, a dependency whose necessity has never been empirically tested. We ask a precise question: is the fusion gain driven by the LLM’s linguistic rewriting, or by the simpler effect of deduplicating and packing verbatim evidence into a fixed retrieval budget? To answer it, we propose DFM-Fusion, a fully deterministic operator requiring zero maintenance-time LLM calls. More concretely, DFM-Fusion applies four sequential stages — sentence segmentation, near-duplicate removal, MMR-based sentence packing, and deterministic coverage verification — to consolidate memory clusters without any generative rewriting. Key to DFM-Fusion’s design is the recognition that verbatim evidence packing, not LLM rewriting, accounts for virtually all of the fusion gain in fact-centric conversational QA. Across six open-weight model families on LoCoMo, DFM-Fusion achieves the best average multi-hop F1 (0.1877) versus LLM-Fusion (0.1841) and No-Fusion (0.1785), improving in 5/6 models while reducing maintenance overhead by $69.13\times$ and eliminating all fusion-specific API calls. The answer to our question is clear: effective memory consolidation for long-horizon conversational agents can be made entirely free of API dependencies in the maintenance loop.

1 INTRODUCTION

Consider a long-running personal assistant that has conversed with a user across hundreds of sessions over months. When the user asks a multi-hop question — “Which hobby did I pick up right after quitting the gym?” — the agent must integrate two distant conversation fragments out of thousands of accumulated memory entries. The challenge is not retrieval quality alone: as related interactions accumulate, a fixed retrieval budget increasingly fills with near-duplicate evidence rather than the distinct facts that such questions demand. This retrieval-noise problem is not hypothetical: on LoCoMo’s multi-hop benchmark, which requires integrating evidence across multiple conversation sessions, even strong systems score below 20 F1 without effective memory consolidation (Maharana et al., 2024).

Memory fusion — the consolidation of semantically similar fragments into compact, non-redundant representations — is the single largest performance driver in state-of-the-art long-horizon memory systems. In FadeMem’s ablation study, removing fusion while holding everything else fixed drops multi-hop F1 from 29.43 to 13.63, a 53.7% collapse (Wei et al., 2026). Fusion is not a peripheral optimization; it is what makes long-horizon conversational QA work at all.

However, every system that performs fusion does so through LLM-guided consolidation: an LLM rewrites the memory cluster into a coherent summary, then a second LLM call verifies that the rewrite preserved information (Wei et al., 2026; Chhikara et al., 2025). This pipeline works, but it rests on an unexamined assumption whose necessity has never been empirically tested. Our key insight is that in fact-centric conversational QA, the fusion gain comes almost entirely from dedupli-

cating and packing verbatim evidence into the fixed retrieval budget — not from the LLM’s linguistic rewriting ability. Different from prior works that treat LLM rewriting as an indispensable step, we show that a fully deterministic operator preserving verbatim sentences recovers the same gain while requiring zero additional LLM calls at maintenance time. This changes how practitioners design agent memory: persistent, consolidating memory becomes deployable without any API dependency in the maintenance loop, making it accessible for local deployments, cost-sensitive applications, and settings where reproducibility or auditability is required.

We propose **DFM-Fusion** (Deterministic FadeMem-Fusion), a four-stage pipeline that replaces LLM-guided fusion with fully deterministic operations. Concretely, given a cluster of semantically similar memory entries, DFM-Fusion applies sentence segmentation, near-duplicate removal via embedding similarity, MMR-based sentence packing within a token budget, and salient-token coverage verification with a fallback guarantee — without invoking any LLM at any stage. We evaluate the same memory pipeline across six open-weight model families on LoCoMo while holding the retrieval budget, scoring protocol, and answer-generation stack fixed. Our contributions are:

- A deterministic, quote-preserving fusion operator that eliminates all LLM calls from memory consolidation while maintaining auditable, verbatim evidence packing.
- A six-model family evaluation on LoCoMo showing that DFM-Fusion achieves the strongest average multi-hop F1 (0.1877) and improves over LLM-Fusion on the primary metric in 5/6 models — confirming that the fusion gain is driven by information packing, not LLM rewriting.
- Consistent operational gains: $69.13\times$ average maintenance speedup, 11.74% reduction in total API calls, and complete elimination of fusion-specific LLM calls — with no performance cost.

The conclusion is deliberately scoped: for fact-centric long-horizon conversational QA, where the evaluation rewards verbatim evidence retrieval, LLM-guided fusion is unnecessary. Whether the same holds for tasks requiring abstraction or conflict reconciliation remains open, and we discuss this boundary explicitly in Section 5.

2 RELATED WORK

Memory Systems for Conversational Agents. Long-horizon conversational agents require persistent memory to maintain coherence across extended interactions. Earlier work established the core architectural patterns: MemGPT (Packer et al., 2023) pioneered hierarchical storage tiers through an operating-system metaphor for LLM context management, while FadeMem (Wei et al., 2026) introduced biologically-inspired forgetting with LLM-guided fusion that consolidates related memories as less relevant ones decay. More recent systems extend this foundation through alternative organizational schemes: Mem0 (Chhikara et al., 2025) captures relational structure via graph-based representations, TiMem (Li et al., 2026) organizes memories through temporal hierarchies, SimpleMem (Liu et al., 2026) achieves efficiency through semantic compression, and EverMemOS (Hu et al., 2026) models engram-inspired memory lifecycles. However, across all of these systems, memory consolidation invariably requires additional LLM calls — for rewriting, merging, or preservation verification — a dependency whose necessity has never been empirically isolated from the simpler effect of redundancy control under a fixed retrieval budget.

Redundancy-Aware Selection in Information Retrieval. The algorithmic foundations for deterministic redundancy control are well established in information retrieval. Maximal Marginal Relevance (Carbonell & Goldstein, 1998) explicitly trades off relevance and diversity during document ranking, and near-duplicate removal via embedding similarity is a standard preprocessing step in retrieval pipelines. However, these techniques operate at query time — selecting among already-stored documents to populate a retrieval result — rather than at maintenance time, where the task is to consolidate redundant memory entries into compact representations before retrieval occurs. Whether their principles transfer directly to memory maintenance, replacing generative rewriting, has not been tested.

Retrieval-Augmented Generation. RAG approaches (Gao et al., 2023) retrieve relevant context to augment LLM generation. HippoRAG (Gutierrez et al., 2024) draws inspiration from hippocampal indexing theory, using knowledge graphs and personalized PageRank for retrieval. GraphRAG (Edge et al., 2024) leverages entity knowledge graphs with community summaries for query-focused summarization. While these methods advance retrieval quality, they assume a static document store and do not address the memory consolidation problem where redundant entries must be continuously merged while preserving information fidelity.

Memory Benchmarks. LoCoMo (Maharana et al., 2024) evaluates long-term conversational memory through multi-hop question answering over 300-turn dialogues spanning 35 sessions. BEAM (Tavakoli et al., 2025) extends evaluation to million-token contexts with diverse memory probing tasks. These benchmarks reveal that even long-context LLMs struggle with extended dialogues, motivating the need for effective memory management — yet neither directly evaluates whether LLM-guided memory consolidation is necessary, or whether equivalent performance can be achieved through deterministic maintenance operators.

3 METHOD

We propose DFM-Fusion (Deterministic FadeMem-Fusion), a fully deterministic replacement for LLM-guided memory fusion in long-horizon conversational agents. Our approach maintains the memory architecture and retrieval mechanisms of existing systems while eliminating all LLM calls from the fusion operator.

3.1 PROBLEM FORMULATION

Memory fusion addresses the challenge of consolidating redundant information in conversational memory stores. Given a cluster of semantically similar memory entries $\mathcal{M} = \{m_1, m_2, \dots, m_k\}$ identified through temporal-semantic clustering, the fusion task produces a single consolidated entry m_f that preserves essential information within a token budget B .

Existing approaches (Wei et al., 2026; Chhikara et al., 2025) employ LLMs to rewrite and merge memories, then use an additional LLM call to verify that the rewrite preserved information. As discussed in Section 1, this introduces deployment challenges related to cost, determinism, and auditability. Our approach addresses these by preserving verbatim spans while enforcing explicit token budgets.

Memory strength. We follow the FadeMem-style strength dynamics used by the implementation. Each new memory starts with strength $v_i = 1.0$ in the short-term layer. At time t , its importance is

$$I_i(t) = \alpha r_i(t) + \beta \frac{a_i}{1 + a_i} + \gamma \exp(-\delta \Delta t_i), \quad (1)$$

where $r_i(t)$ is cosine relevance to the current session context when available, a_i is access frequency, and Δt_i is age in days. Strength decays as

$$v_i(t) = v_i(t_0) \exp[-\lambda_{\text{base}} \exp(-\mu I_i(t)) (\Delta t_i)^\rho], \quad (2)$$

with $\rho = 0.8$ for long-term memories and $\rho = 1.2$ for short-term memories. Access consolidation increases strength by

$$v_i \leftarrow v_i + d_v(1 - v_i) \exp(-c_i/N), \quad (3)$$

where c_i is access count. A fused memory receives strength

$$v_f = \min\left(1, \max_i v_i + \epsilon \text{Var}(\{v_i\})\right), \quad (4)$$

and is inserted into the long-term layer. In our experiments $\alpha = 0.4$, $\beta = 0.3$, $\gamma = 0.3$, $\delta = 0.1$, $\mu = 2.0$, $d_v = 0.15$, $N = 5$, and $\epsilon = 0.1$.

3.2 DFM-FUSION PIPELINE

Given a cluster $\mathcal{M}$ exceeding a minimum size threshold, DFM-Fusion produces fused text s_f through four deterministic stages. More specifically: (1) sentence segmentation breaks each memory entry into sentence-level units to enable fine-grained selection; (2) near-duplicate removal eliminates redundant sentences via embedding similarity; (3) MMR-based sentence packing selects a diverse, high-strength subset within the token budget; and (4) coverage verification confirms that salient evidence is preserved, with a fallback guarantee if it is not. No LLM is invoked at any stage.

Stage 1: Sentence Segmentation. Each memory entry m_i is split into individual sentences using rule-based tokenization. However, applying fusion directly at the memory-entry level would truncate content mid-sentence when the token budget is enforced, fragmenting the verbatim spans that downstream retrieval and F1 evaluation depend on. Sentence-level segmentation ensures that every selected unit is a complete, citable span.

Stage 2: Near-Duplicate Removal. We compute pairwise cosine similarity between sentence embeddings using Sentence-BERT (Reimers & Gurevych, 2019) (384-dim). However, the raw sentence pool from a cluster frequently contains near-identical reformulations of the same fact — variants that would consume token budget without adding evidence coverage. Sentences with cosine similarity exceeding threshold $\theta_{\text{dup}} = 0.85$ are treated as near-duplicates; we retain the sentence from the memory with higher strength $v_i(t)$, using a more recent timestamp as a tie-breaker. The same Sentence-BERT encoder is reused throughout the pipeline — for deduplication, for MMR scoring, and for computing the retrieval embedding of the final fused entry — eliminating any inconsistency between maintenance-time and retrieval-time representations.

Stage 3: MMR-Based Sentence Packing. From the deduplicated sentence pool, we select sentences using a strength-aware variant of Maximal Marginal Relevance (Carbonell & Goldstein, 1998) with diversity parameter $\lambda = 0.7$. However, relevance scoring alone would fill the budget with the highest-strength sentences from the dominant source memory, crowding out lower-strength entries that may contain complementary evidence. The MMR objective balances source-memory strength against redundancy with already-selected sentences:

$$\text{MMR}(s_i) = \lambda \cdot \hat{v}(s_i) - (1 - \lambda) \cdot \max_{s_j \in S} \cos(s_i, s_j) \quad (5)$$

where $\hat{v}(s_i) \in [0, 1]$ is the normalized strength of the source memory for sentence s_i , and S is the set of already-selected sentences. We greedily pick $\arg \max_{s_i \in R \setminus S} \text{MMR}(s_i)$ until adding the next sentence would exceed the token budget $B_{\text{fuse}} = 768$ tokens. Selected sentences are sorted by source-memory timestamp before concatenation. The fused memory embedding is computed from the actual fused text s_f , not by averaging constituent embeddings; this ensures retrieval operates on a representation that accurately reflects the consolidated content, and was validated by our ablation (Table 1, lower block).

Stage 4: Coverage Verification. To confirm information preservation without an LLM, we implement a deterministic coverage check. We extract salient evidence from the original cluster: numbers and date-like expressions, capitalized entities, top- K TF-IDF weighted tokens ($K = 20$), and stemmed n -gram coverage features in the enriched configuration. Coverage recall is the fraction of these tokens appearing in s_f . It should be noted that a coverage gate that simply rejects low-recall fusions would leave clusters unmerged, breaking the consolidation guarantee. To address this, when recall falls below threshold $\theta_{\text{cov}} = 0.85$, we fall back to concatenating the highest-strength original memories within the budget in strength-descending order, so a fused result is always emitted rather than leaving the cluster unmerged.

3.3 HYPERPARAMETER SELECTION

FadeMem decay parameters ($\theta_{\text{fusion}} = 0.75$, $T_{\text{window}} = 10$, $\lambda_{\text{base}} = 0.1$) are adopted directly from the original system. DFM-specific fusion parameters were initialized to interpretable defaults ($\theta_{\text{dup}} = 0.90$, $B_{\text{fuse}} = 512$, $\lambda = 0.5$, $K = 10$, $\theta_{\text{cov}} = 0.80$) and revised once, in a single diagnostic pass that identified four concrete issues: an embedding representation bug (weighted-average rather than fused-text encoding), a suboptimal text separator, an over-aggressive deduplication threshold,

and an undersized fusion budget. This pass was conducted on the full LoCoMo-10 evaluation set without a held-out split. While the changes are individually motivated by interpretable diagnostics rather than metric search, we acknowledge that we cannot fully rule out mild overfitting to the 10 evaluation conversations; the six-model cross-family replication (Section 4) provides indirect stability evidence.

4 EXPERIMENTS

We evaluate DFM-Fusion on LoCoMo to test whether deterministic fusion can replace LLM-guided fusion as the default memory-maintenance operator under a fixed retrieval and answer-generation pipeline.

4.1 EXPERIMENTAL SETUP

Dataset. We use LoCoMo-10 (Maharana et al., 2024), a subset of the LoCoMo benchmark containing 10 conversations with 1,986 QA pairs. Questions span five categories: multi-hop (282), single-hop (817), temporal (321), open-domain (96), and adversarial (446). Multi-hop F1 serves as our primary metric, as it requires integrating information across multiple memory entries and is most sensitive to fusion quality.

Cross-model protocol. The main empirical evaluation is a six-model family study. We run the same memory pipeline across six open-weight answer-generation models served locally via Ollama: qwen25-7b, qwen25-14b, qwen3-8b, llama31-8b, llama32-3b, and mistral-7b. The answer-generation LLM is the sole experimental variable; the memory pipeline (Sentence-BERT embeddings (Reimers & Gurevych, 2019), identical retrieval budgets, and DFM-Fusion or LLM-Fusion operator) is held constant across models. LLM-Fusion maintenance calls (fusion generation and preservation checking) always use DeepSeek-V3.2 regardless of which local answer model runs, replicating FadeMem’s API-based maintenance design. For each model we compare three conditions: (1) **LLM-Fusion**: FadeMem-style fusion with additional DeepSeek-V3.2 calls for fusion generation and preservation checking; (2) **DFM-Fusion**: our deterministic fusion operator with no maintenance-time LLM calls; and (3) **No-Fusion**: no memory merging. All conditions share identical retrieval budgets, scoring protocol, and memory infrastructure.

An additional single-model pilot using DeepSeek-V3.2 as the answer model (3 runs per condition, temperature=0) validated the harness and is reported in Appendix A (Table 7) as a sanity check only; it is not part of the main results.

Statistical tests. We report mean $\pm$ std over 3 runs per condition. Our primary paired comparison is DFM-Fusion vs. LLM-Fusion on per-question multi-hop F1 scores. We report mean differences, paired t-test p -values, and TOST equivalence outcomes with a pre-specified equivalence margin of ± 0.01 absolute multi-hop F1. This margin corresponds to a practically negligible change relative to the observed No-Fusion to LLM-Fusion gaps on LoCoMo-10. The resulting evidence should be interpreted conservatively: only one model satisfies the configured TOST equivalence criterion, so our main claim is practical competitiveness rather than universal statistical equivalence.

4.2 MAIN RESULTS

Accuracy results. Table 1 shows that DFM-Fusion improves over LLM-Fusion on multi-hop F1 in 5/6 models; the only exception is Qwen2.5-7B, where the gap is negligible (-0.0007). The strongest individual result is Qwen3-8B, where DFM-Fusion reaches 0.2157, outperforming LLM-Fusion by $+0.0100$. Averaged across the six families, DFM-Fusion leads by $+0.0036$ on multi-hop F1. The key driver of this advantage is verbatim evidence packing within the retrieval budget, not LLM rewriting — a mechanism that operates identically across model families, which is why the directional result holds consistently despite the heterogeneity of the six answer models.

We observed that the per-category breakdown makes the scope of this advantage precise. DFM-Fusion leads on multi-hop and single-hop categories, where success rewards redundancy-aware packing of verbatim evidence. LLM-Fusion retains small advantages on temporal, open-domain,

Table 1: Comprehensive results on LoCoMo-10. Per-model rows show DFM-Fusion (DFM), LLM-Fusion (LLM), and No-Fusion (No) across all six question categories; Δ is DFM minus LLM on multi-hop F1; p is the paired t-test p -value. The **Average** row reports six-model means; bolded cells indicate the best-performing method per column. Ablation rows (*indented*) show DFM variant averages; *italicized* LLM and No values repeat the fixed baselines from the Average row (LLM-Fusion and No-Fusion are unaffected by DFM ablations); $\Delta^\dagger$ compares the ablation to DFM-Fusion on multi-hop F1. DFM-Fusion eliminates all maintenance-time LLM calls: $69.13\times$ average maintenance speedup, 11.74% total-call reduction, 3.88% cost savings. Cost figures use DeepSeek-V3.2 pricing (\$0.27/1M input, \$1.10/1M output); local Ollama models incur zero direct API cost.

Model / Method	Multi-hop F1			Overall F1			Temporal F1			Open-domain F1			Single-hop F1			Adversarial F1			Δ	p
	DFM	LLM	No	DFM	LLM	No	DFM	LLM	No	DFM	LLM	No	DFM	LLM	No	DFM	LLM	No		
Qwen2.5-7B	0.1832	0.1839	0.1769	0.3703	0.3709	0.3644	0.0536	0.0588	0.0536	0.0792	0.0785	0.0739	0.3269	0.3238	0.3133	0.8610	0.8655	0.8655	-0.0007	0.9142
Qwen2.5-14B	0.1836	0.1833	0.1697	0.3714	0.3728	0.3682	0.0505	0.0531	0.0538	0.1216	0.1217	0.1170	0.2830	0.2841	0.2817	0.9417	0.9439	0.9372	+0.0003	0.9510
Qwen3-8B	0.2157	0.2057	0.1992	0.3683	0.3663	0.3643	0.0575	0.0619	0.0610	0.1229	0.1274	0.1394	0.3334	0.3317	0.3262	0.8072	0.8034	0.8072	+0.0100	0.1493
Llama3.1-8B	0.1909	0.1881	0.1825	0.3674	0.3650	0.3611	0.0566	0.0560	0.0580	0.1030	0.1092	0.1108	0.3137	0.3079	0.3085	0.8610	0.8617	0.8453	+0.0028	0.5896
Llama3.2-3B	0.1795	0.1737	0.1780	0.3348	0.3321	0.3337	0.0536	0.0534	0.0494	0.0941	0.0983	0.0838	0.2676	0.2640	0.2694	0.8139	0.8117	0.8117	+0.0058	0.0698
Mistral-7B	0.1731	0.1699	0.1644	0.3383	0.3395	0.3354	0.0549	0.0596	0.0599	0.1046	0.1047	0.1015	0.2907	0.2867	0.2852	0.7870	0.7982	0.7870	+0.0032	0.5526
Average	0.1877	0.1841	0.1785	0.3584	0.3577	0.3545	0.0545	0.0571	0.0560	0.1042	0.1066	0.1044	0.3026	0.2997	0.2974	0.8453	0.8474	0.8423	+0.0036	—
DFM w/o budget truncation	0.1875	<i>0.1841</i>	<i>0.1785</i>	0.3584	0.3577	0.3545	0.0545	0.0571	0.0560	0.1042	0.1066	0.1044	0.3025	0.2997	0.2974	0.8453	<i>0.8474</i>	<i>0.8423</i>	-0.0002 [†]	—
DFM w/ basic coverage	0.1871	<i>0.1841</i>	0.1785	0.3580	0.3577	0.3545	0.0545	0.0571	0.0560	0.1037	0.1066	0.1044	0.3018	0.2997	0.2974	0.8453	<i>0.8474</i>	<i>0.8423</i>	-0.0006 [†]	—
DFM w/o coverage gate	0.1867	<i>0.1841</i>	0.1785	0.3581	0.3577	0.3545	0.0557	0.0571	0.0560	0.1042	0.1066	0.1044	0.3008	0.2997	0.2974	0.8468	<i>0.8474</i>	<i>0.8423</i>	-0.0010 [†]	—
DFM w/ mean embedding	0.1826	<i>0.1841</i>	0.1785	0.3577	0.3577	0.3545	0.0579	0.0571	0.0560	0.1035	0.1066	0.1044	0.3009	0.2997	0.2974	0.8460	<i>0.8474</i>	<i>0.8423</i>	-0.0051 [†]	—

and adversarial categories, where success requires paraphrastic synthesis or coarse temporal normalization — capabilities that verbatim packing cannot replicate. To illustrate where this boundary lies: on a temporal question such as “When did Caroline join a new activist group?”, DFM-Fusion retrieved topically relevant evidence but could not normalize it into the expected relative temporal expression, returning no date; LLM-Fusion produced a coarse but partially correct relative date. This failure mode is consistent with the average-row temporal gap (DFM 0.0545 vs. LLM 0.0571) and marks exactly the category where generative rewriting retains value.

Among the ablation variants, DFM w/ mean embedding is the weakest on multi-hop F1 (-0.0051 vs. DFM-Fusion) while showing the highest temporal F1 (0.0579). We observed that this reversal is consistent with averaged embeddings over-smoothing local structure: mean embeddings retrieve broader, more topically diffuse evidence that occasionally matches temporal queries but fails on the precise multi-hop integration that exact-text embeddings support. The remaining three ablations stay within 0.0010 of DFM-Fusion on multi-hop F1 and differ negligibly across all categories.

Retrieval and fusion diagnostics. Retrieval-side diagnostics support the same direction. Averaged across six models, DFM-Fusion yields slightly stronger evidence recall / hit rate / MRR (0.3667 / 0.4105 / 0.2067) than LLM-Fusion (0.3662 / 0.4098 / 0.2064) and No-Fusion (0.3628 / 0.4070 / 0.2045). Each DFM fusion event packs on average 9.65 selected sentences into 106.64 tokens, with mean coverage of 0.9189 for capitalized entities, 0.8278 for TF-IDF salient terms, 0.8862 for stemmed n -grams, and 1.0 for both dates and numeric tokens. Retrieval diagnostics and fusion behavior statistics (Tables 3–4) are identical or near-identical across the six answer models by design: these quantities depend only on the memory pipeline and the shared Sentence-BERT encoder (Reimers & Gurevych, 2019), not on which local model generates answers.

Operational efficiency. DFM-Fusion eliminates all maintenance-time LLM calls, replacing them with CPU-only deterministic operations. This yields a $69.13\times$ average maintenance speedup, an 11.74% reduction in total API calls, and a 3.88% reduction in cost — at no accuracy cost on the primary metric. Per-model operational breakdowns, retrieval diagnostics, paired tests, TOST outcomes, and targeted error analysis appear in the appendix.

5 DISCUSSION AND LIMITATIONS

No universal equivalence claim. The revised evidence does not support a blanket statement that DFM-Fusion is statistically equivalent to LLM-Fusion everywhere. Under the configured TOST test, only Qwen2.5-14B satisfies equivalence. The correct claim is therefore practical rather than absolute: DFM-Fusion is competitive or better on the primary multi-hop metric across a heterogeneous model family, with consistent operational savings.

Benchmark scope. All results are on LoCoMo, which is an appropriate benchmark for fact-centric long-horizon conversational QA but not a sufficient basis for claiming that deterministic fusion replaces LLM-guided consolidation in every setting. The six-model cross-family replication provides evidence of robustness across answer-model families, but not across benchmarks.

Serving-stack dependence. Quality results are model-level, but latency and maintenance numbers are partly serving-stack dependent. The exact speedup factors reported here should therefore be interpreted as deployment-relevant evidence rather than universal constants. What appears robust is the directional result: eliminating maintenance-time fusion calls substantially simplifies and accelerates the memory-maintenance loop.

Surface-biased preservation heuristics. Our deterministic preservation check is based on surface proxies rather than deep semantic entailment. The enriched configuration captures capitalized entities, dates and numerics, TF-IDF salient tokens, and stemmed n -gram coverage, yet it remains English- and formatting-dependent. This leaves open failure modes involving implicit paraphrases, noisy conversational logs, or multilingual settings.

Fallback asymmetry with LLM-Fusion. DFM-Fusion’s coverage gate, when triggered, falls back to concatenating the highest-strength originals within budget — a fused memory is always produced. LLM-Fusion has no equivalent fallback: a failed preservation check discards the entire cluster, leaving memories unmerged. In practice this asymmetry is moot for the reported experiments (LLM-Fusion rejected 0/113 clusters across all six models), but it means DFM-Fusion is more conservative at the preservation gate while still guaranteeing a consolidation result.

Embedding model sensitivity. All reported experiments use Sentence-BERT (Reimers & Gurevych, 2019) for both sentence deduplication and retrieval. Cross-encoder sensitivity was not part of the main result. We provide runnable reviewer-study configurations for E5 and BGE-style sentence encoders, but treat those as sensitivity checks rather than a new headline claim.

Single-benchmark scope and benchmark generalization. All results are on LoCoMo, which is appropriate for fact-centric long-horizon conversational QA. Tasks requiring abstraction, preference synthesis, or explicit reconciliation of conflicting evidence may benefit more from generative maintenance operators, as the targeted error analysis already shows (temporal normalization, open-domain preference synthesis). Evaluation on additional benchmarks such as LongMemEval is an important next step.

Query-time redundancy control. The main paper compares maintenance-time fusion operators. It does not yet fully disentangle maintenance-time consolidation from query-time redundancy control. To address this directly, we implemented additional reviewer-study baselines that apply query-time MMR and query-time deduplication without maintenance fusion under the same retrieval budget. These baselines test whether the observed gains come from persistent memory consolidation or simply from diversity-aware retrieval at inference time.

Why gap recovery is no longer the right headline. In the earlier single-setting draft, gap recovery was a convenient scalar because the ordering between No-Fusion and LLM-Fusion was clean. In the six-model family this is less stable; for example, Llama3.2-3B places No-Fusion slightly above LLM-Fusion on multi-hop F1. For this reason, we emphasize direct DFM–LLM deltas, model-count wins (5/6), and cross-method averages rather than using gap recovery as the primary summary statistic.

6 CONCLUSION

The central finding of this paper is that the gain attributed to LLM-guided memory fusion in fact-centric long-horizon conversational QA comes almost entirely from deduplicating and packing verbatim evidence into a fixed retrieval budget — not from the LLM’s linguistic rewriting ability. DFM-Fusion, a fully deterministic four-stage operator requiring zero maintenance-time LLM calls, recovers this gain across a heterogeneous family of six open-weight models, confirming that effective

memory consolidation can be made entirely free of API dependencies in the maintenance loop. For practitioners, this means that persistent, consolidating memory is deployable in local, cost-sensitive, and reproducibility-constrained settings without any sacrifice in multi-hop reasoning performance.

At the same time, our results do not support a blanket equivalence claim. DFM-Fusion’s advantage is specific to fact-centric QA, where the evaluation rewards verbatim evidence retrieval; LLM-Fusion retains advantages on temporal normalization and open-domain preference synthesis, where paraphrastic rewriting adds genuine value. Future work should extend this investigation to benchmarks requiring abstraction and conflict reconciliation, and explore hybrid operators that apply deterministic packing by default while invoking selective LLM rewriting only when coverage verification signals a genuine synthesis gap. Limitations of the current study — including single-benchmark scope, surface-biased preservation heuristics, and the absence of a held-out validation split — are discussed in Section 5.

A ADDITIONAL RESULTS

A.1 PER-MODEL OPERATIONAL BREAKDOWN

Table 2: Per-model latency, call volume, and cost breakdown. Cost figures use DeepSeek-V3.2 pricing (\$0.27/1M input, \$1.10/1M output) applied to all LLM calls. Local Ollama answer models incur zero direct API cost; their cost figures reflect maintenance-call overhead only. DFM-Fusion cost is identical across all six models (zero fusion calls, same QA call count). Answer-generation cost estimated at 1,585 input + 30 output tokens per QA call.

Model	LLM calls	DFM calls	LLM fusion calls	LLM maint.	DFM maint.	LLM total	DFM total	LLM cost	DFM cost
Qwen2.5-7B	2212	1986	226	332.6	5.5	458.2	272.2	0.948	0.916
Qwen2.5-14B	2292	1986	306	635.0	5.7	1018.4	705.3	0.960	0.916
Qwen3-8B	2212	1986	226	423.0	9.5	618.2	384.2	0.948	0.916
Llama3.1-8B	2264	1986	278	473.3	4.3	893.8	637.2	0.955	0.916
Llama3.2-3B	2280	1986	294	393.4	7.7	618.2	456.9	0.957	0.916
Mistral-7B	2244	1986	258	296.8	7.6	695.7	542.0	0.952	0.916

A.2 RETRIEVAL DIAGNOSTICS

Table 3: Per-model retrieval quality diagnostics. Retrieval statistics depend solely on the memory pipeline and the shared Sentence-BERT encoder (Reimers & Gurevych, 2019), not on the answer-generation model; identical or near-identical values across the six answer models are expected by design.

Model	DFM Rec.	LLM Rec.	No Rec.	DFM Hit	LLM Hit	No Hit	DFM MRR	LLM MRR	No MRR
Qwen2.5-7B	0.3667	0.3657	0.3628	0.4105	0.4093	0.4070	0.2067	0.2063	0.2045
Qwen2.5-14B	0.3667	0.3667	0.3628	0.4105	0.4103	0.4070	0.2067	0.2067	0.2045
Qwen3-8B	0.3667	0.3657	0.3628	0.4105	0.4093	0.4070	0.2067	0.2063	0.2045
Llama3.1-8B	0.3667	0.3676	0.3628	0.4105	0.4112	0.4070	0.2067	0.2066	0.2045
Llama3.2-3B	0.3667	0.3657	0.3628	0.4105	0.4093	0.4070	0.2067	0.2064	0.2045
Mistral-7B	0.3667	0.3657	0.3628	0.4105	0.4093	0.4070	0.2067	0.2063	0.2045

A.3 FUSION BEHAVIOR

A.4 TOST OUTCOMES

A.5 TARGETED ERROR ANALYSIS

Across the inspected model logs, DFM-Fusion underperforms LLM-Fusion in two recurring ways. The dominant failure mode is temporal retrieval miss: the retrieved evidence is either absent or incomplete, after which LLM-Fusion sometimes still answers with a coarse relative date such as “last Tuesday” or “last week.” A second, smaller failure mode appears in open-domain questions that require paraphrastic synthesis rather than verbatim evidence packing. These patterns match the category-level averages in Table 1 (lower block), where LLM-Fusion retains small advantages on temporal and open-domain questions.

Table 4: Per-model DFM-Fusion behavior and coverage diagnostics. DFM-Fusion is fully deterministic; fusion behavior statistics are independent of the answer-generation model and identical across models sharing the same fusion-event history.

Model	Cluster	Selected	Tokens	Fallback	Cap.	TF-IDF	Stem	Date	Num.
Qwen2.5-7B	3.4592	9.6735	106.8878	46	0.9189	0.8301	0.8852	1.0000	1.0000
Qwen2.5-14B	3.4592	9.6735	106.8878	46	0.9189	0.8301	0.8852	1.0000	1.0000
Qwen3-8B	3.4490	9.6224	106.3878	44	0.9189	0.8255	0.8872	1.0000	1.0000
Llama3.1-8B	3.4592	9.6735	106.8878	46	0.9189	0.8301	0.8852	1.0000	1.0000
Llama3.2-3B	3.4490	9.6224	106.3878	44	0.9189	0.8255	0.8872	1.0000	1.0000
Mistral-7B	3.4490	9.6224	106.3878	44	0.9189	0.8255	0.8872	1.0000	1.0000

Table 5: Per-model TOST equivalence outcomes for DFM-Fusion vs. LLM-Fusion on multi-hop F1.

Model	DFM-LLM delta	<i>p</i> -value	TOST equivalent
Qwen2.5-7B	-0.0007	0.9142	false
Qwen2.5-14B	+0.0003	0.9510	true
Qwen3-8B	+0.0100	0.1493	false
Llama3.1-8B	+0.0028	0.5896	false
Llama3.2-3B	+0.0058	0.0698	false
Mistral-7B	+0.0032	0.5526	false

A.6 DEEPSEEK-V3.2 SINGLE-MODEL SANITY CHECK

Before running the six-model family study, we validated the harness using DeepSeek-V3.2 (API, temperature=0) as the answer model, with 3 independent runs per condition. These results confirmed the directional ordering (DFM-Fusion > No-Fusion, DFM-Fusion $\approx$ LLM-Fusion) and are included only as a harness sanity check; the main empirical basis of the paper is the six-model local-model evaluation in Section 4.

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Table 6: Representative targeted error cases extracted from model-level analyses.

Model	Failure mode	Representative question	Interpretation
Qwen2.5-7B	answer generation mismatch	Which pet did Jolene adopt first, Susie or Seraphim?	Retrieved evidence was partial, and DFM answered “Not mentioned” while LLM-Fusion returned the correct short answer.
Qwen3-8B	temporal retrieval miss	When did Caroline join a new activist group?	DFM returned no date after a retrieval miss, while LLM-Fusion produced a coarse but partially correct relative date.
Qwen3-8B	temporal normalization	When did Evan meet his future wife?	DFM preserved topical evidence (Canada trip) but failed to normalize it into the expected relative temporal expression.
Llama3.1-8B	paraphrase synthesis or	Which type of vacation would Evan prefer, city walking tours or camping outdoors?	The open-domain answer requires synthesizing preference evidence into a short decision, a setting where LLM-Fusion still has an advantage.

Table 7: DeepSeek-V3.2 single-model sanity check (3 runs per condition, temperature=0). Reported as harness validation only.

Setting	Multi-hop F1	Overall F1
LLM-Fusion	0.1863 $\pm$ 0.0021	0.3805 $\pm$ 0.0004
DFM-Fusion	0.1872 $\pm$ 0.0034	0.3795 $\pm$ 0.0012
No-Fusion	0.1710 $\pm$ 0.0009	0.3750 $\pm$ 0.0011

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